

Aman Kumar

+91-7061052141 | kumaramansingh2005@gmail.com | [linkedin.com/in/aman-kumar-518a18304](https://www.linkedin.com/in/aman-kumar-518a18304)
| <https://github.com/amandevspace>

PROFILE

Aspiring **Software Engineer** with strong focus on **MERN stack, backend systems, and scalable distributed architectures**. Experienced in building **high-performance APIs, real-time applications, and AI-integrated platforms** with emphasis on performance, concurrency, and system reliability.

TECHNICAL SKILLS

- **Languages:** Java, JavaScript
- **Frontend:** React.js
- **Backend:** Node.js, Express.js, FastAPI
- **Databases:** MongoDB, Redis
- **Core Concepts:** REST APIs, System Design, Concurrency, Caching, Rate Limiting
- **AI/ML:** LLMs, RAG, Pinecone
- **Tools:** Git, GitHub, Postman, Vercel, Render

PROJECTS

FlowLimiter — API Rate Limiting & Traffic Control System

- Engineered a full-stack rate limiting system using **MERN stack with Redis-backed atomic enforcement**.
- Designed API key management, request tracking, and real-time traffic monitoring with **React dashboard and MongoDB logging**.
- Implemented and optimized **Fixed Window, Token Bucket, and Sliding Window Counter algorithms**.
- Ensured concurrency safety using **atomic Redis operations (INCR, EXPIRE, Lua scripting)** across distributed systems.
- Achieved **6600+ requests/sec throughput with p99 latency of 21 ms**, handling **66K+ requests in 10 seconds**.

APIVigil — AI-Powered API Observability Platform

- Engineered a scalable API observability platform handling **1000+ concurrent API requests**, improving failure detection using **ML-based anomaly detection**.
- Designed and integrated a **RAG pipeline with Pinecone** for intelligent log retrieval and root-cause analysis.
- Developed real-time dashboards and alerting system, reducing issue response time.
- Implemented predictive analytics using **Scikit-learn** to identify abnormal API behavior patterns.

EDUCATION

B.Tech in Computer Science and Engineering

2023 – 2027

Oriental Institute of Science and Technology, Bhopal

CGPA: 7.72

ACHIEVEMENTS

- NPTEL Elite + Topper (Top 5%) in Programming in Java with 96% score
- Published research paper on sign language recognition in IJRPR journal
- Completed AI/ML Virtual Internship (Google for Developers)